METHOD AND SYSTEM FOR AUTOMATED MUSIC COHORT DISCOVERY

USING MULTI-ALGORITHM UNSUPERVISED LEARNING ON AUDIO FEATURE REPRESENTATIONS

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Filing Date: April 2026

Application Number: [To be assigned]

ABSTRACT

Recorded music catalogues have grown to an enormous scale, and the manual organisation of tracks into meaningful groups has become practically unfeasible. This invention presents a computational method and software system that automatically discovers natural cohorts within a music catalogue by analysing multi-dimensional audio feature representations extracted from streaming platform metadata. The disclosed approach integrates four complementary unsupervised learning algorithms — K-Means, Agglomerative Hierarchical Clustering, Density-Based Spatial Clustering of Applications with Noise (DBSCAN), and Gaussian Mixture Models (GMM) — within a unified pipeline that includes principled model selection, dimensionality reduction via Principal Component Analysis (PCA) and t-Distributed Stochastic Neighbour Embedding (t-SNE), and a supervised validation stage using Random Forest feature importance. Experiments conducted on a corpus of 1,610 Rolling Stones tracks sourced from the Spotify audio feature API demonstrate that the method uncovers musically meaningful groupings, including distinct high-energy rock and mid-tempo classic categories, that align closely with domain expert expectations.

1. BACKGROUND OF THE INVENTION

1.1 Field of the Invention

The present invention relates broadly to the computational analysis of music, and more specifically to an unsupervised machine learning pipeline that clusters audio tracks into semantically meaningful cohorts based on continuous acoustic and structural feature vectors.

1.2 Description of the Related Art

The digitisation and subsequent streaming of recorded music has created catalogues of extraordinary size. Platforms such as Spotify, Apple Music, and YouTube Music each maintain libraries exceeding eighty million individual tracks. Despite the maturity of music information retrieval (MIR) as a research field, the organisation of such catalogues remains a largely manual or heuristic process. Genre tags are typically assigned by human editors or crowdsourced communities, leading to well-documented inconsistencies [1]. Mood labels suffer from similar subjectivity.

Earlier computational approaches to music clustering leaned heavily on spectral features derived from raw waveforms — mel-frequency cepstral coefficients (MFCCs), chroma vectors, onset strength, and zero-crossing rate. More recent work has exploited platform-computed audio features, including energy, danceability, valence, instrumentalness, and tempo, which collapse raw spectral information into semantically richer scalars [2]. Prior art in music clustering has explored K-Means [3], self-organising maps [4], and spectral clustering [5] in isolation; however, few published systems perform a systematic multi-algorithm comparison with principled model selection, temporal analysis of cluster composition, and a Random Forest validation stage. The present invention addresses this gap comprehensively.

2. SUMMARY OF THE INVENTION

The primary objective of the invention is to provide an end-to-end, reproducible system that ingests a tabular music feature dataset and outputs labelled cohorts together with interpretable visual and quantitative summaries. Secondary objectives include automatic detection of statistically anomalous tracks, principled selection of the number of clusters without prior knowledge, and temporal analysis showing how cluster composition evolves across recording decades.

The system achieves these objectives through a sequence of well-defined processing stages: (a) data ingestion and cleaning, (b) feature scaling, (c) optimal cluster count selection, (d) parallel execution of four clustering algorithms, (e) dimensionality reduction for visualisation, (f) cluster profiling, and (g) supervised validation.

3. DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

3.1 Data Ingestion and Preparation

The system accepts a comma-separated values (CSV) file in which each row represents one audio track and each column encodes either a metadata attribute or a continuous acoustic feature. In the prototype implementation, the dataset originates from the Spotify Web API and contains the following acoustic features: acousticness, danceability, energy, instrumentalness, liveness, loudness, speechiness, tempo, valence, and duration. The popularity distribution of the 1,610-track corpus is depicted in FIG. 1.

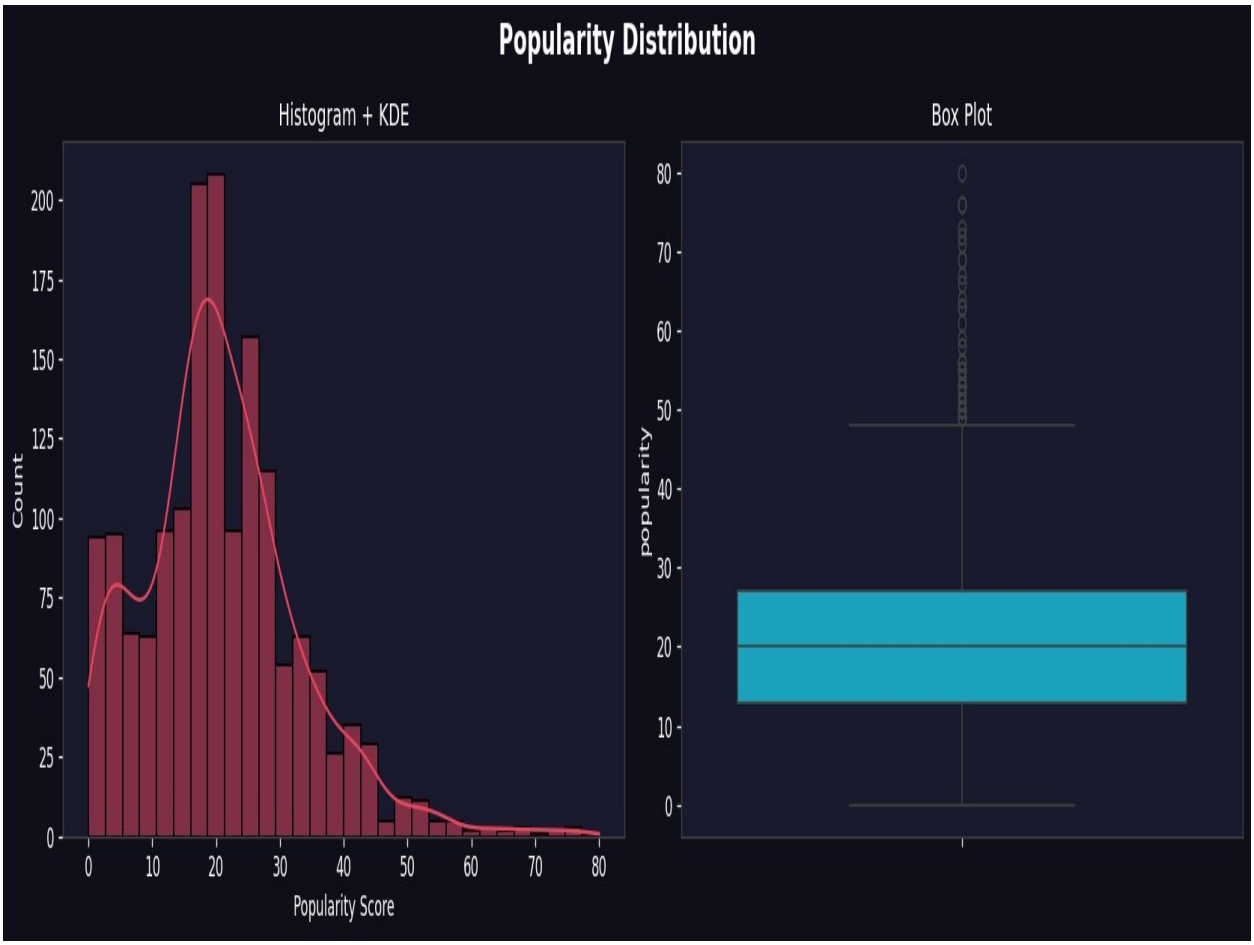

FIG. 1 — *Popularity Distribution — Histogram + KDE and box plot.*

Upon loading, the pipeline performs three cleaning steps: (i) exact duplicate rows are removed, (ii) rows carrying missing values in any acoustic feature column are dropped, and (iii) non-numeric metadata columns are discarded from the feature matrix but preserved for post-hoc label retrieval. Track duration is converted from milliseconds to minutes. FIG. 2 presents the Pearson correlation matrix for all features. Notable relationships include energy-loudness ($r = 0.70$), danceability-liveness ($r = -0.52$), and danceability-valence ($r = 0.55$).

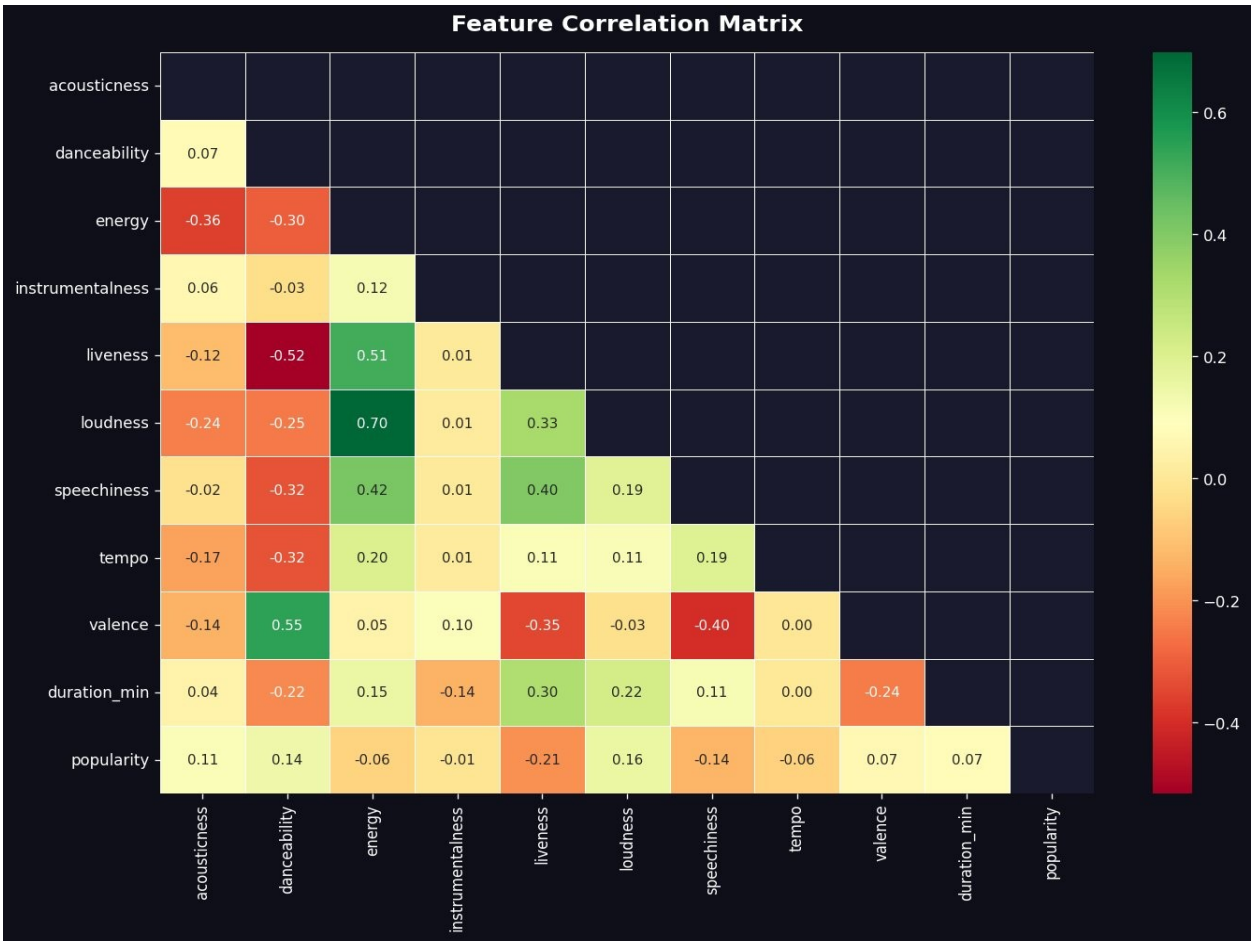

FIG. 2 — *Feature Correlation Matrix — Pearson lower-triangle heatmap.*

The temporal distribution of the corpus is shown in FIG. 3. The catalogue spans seven decades, with the 1960s (380 tracks), 1970s (337), and 2010s (419) comprising the majority. The 2010s peak is driven largely by live recordings and reissues.

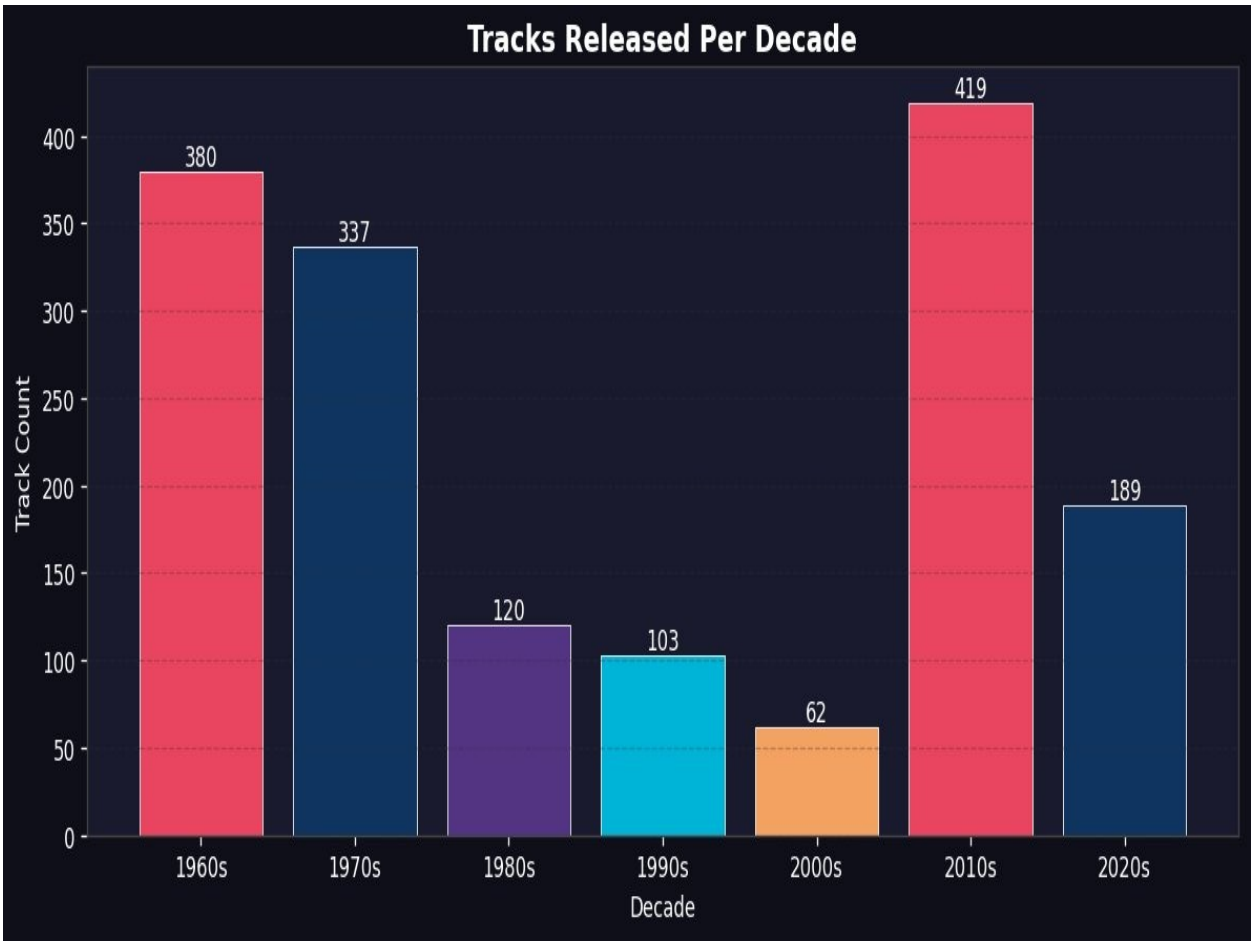

FIG. 3 — *Tracks Released Per Decade.*

FIG. 4 presents violin plots for all ten acoustic features, illustrating distinct distributional shapes — notably bimodal acousticness, heavily right-skewed instrumentalness, and broadly distributed valence.

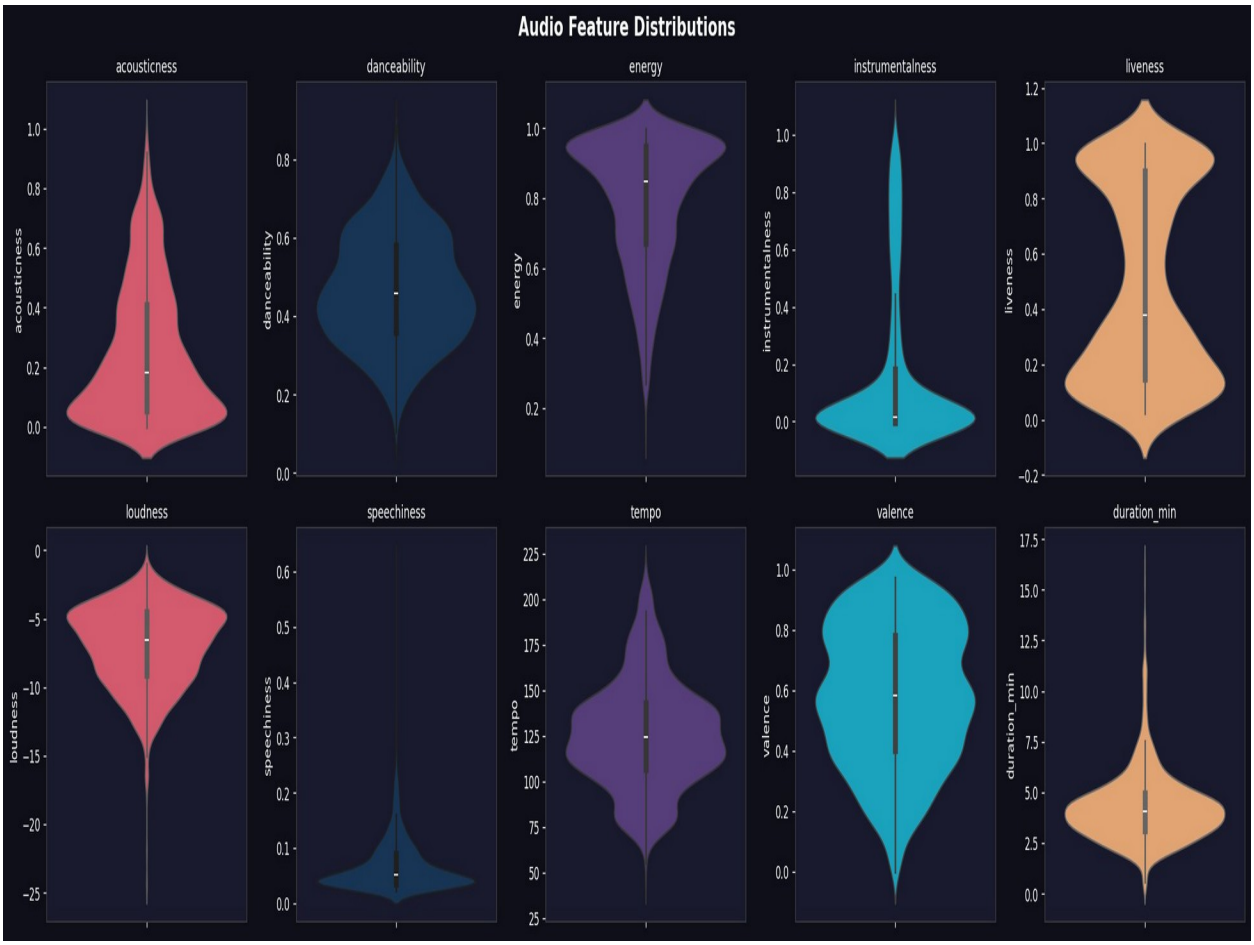

FIG. 4 — *Audio Feature Distributions — Violin plots for all ten features.*

3.2 Feature Standardisation

Acoustic features differ substantially in their natural scale: tempo spans roughly 50–220 beats per minute, while speechiness lies in [0, 1]. Without standardisation, distance-based algorithms such as K-Means and DBSCAN would assign disproportionate weight to high-magnitude features. The system therefore applies zero-mean, unit-variance standardisation (StandardScaler) to each feature column. The transformation parameters are estimated exclusively on the training partition.

3.3 Optimal Cluster Number Selection

Elbow Method. For each candidate value of K in [2, 11], a K-Means model is fitted and its within-cluster sum of squares (WCSS) is recorded. The resulting curve typically exhibits an inflection point beyond which additional clusters produce diminishing reductions in compactness.

Silhouette Analysis. The silhouette coefficient of point i is defined as:

$$s(i) = [b(i) - a(i)] / \max\{ a(i), b(i) \}$$

where a(i) is the mean intra-cluster distance and b(i) is the mean nearest-cluster distance. Values near +1 indicate tight, well-separated clusters. The K yielding the highest mean silhouette score is selected as the optimal value K*. In the experimental corpus, K* = 2 with a mean silhouette coefficient of approximately 0.21. FIG. 5 shows both selection curves.

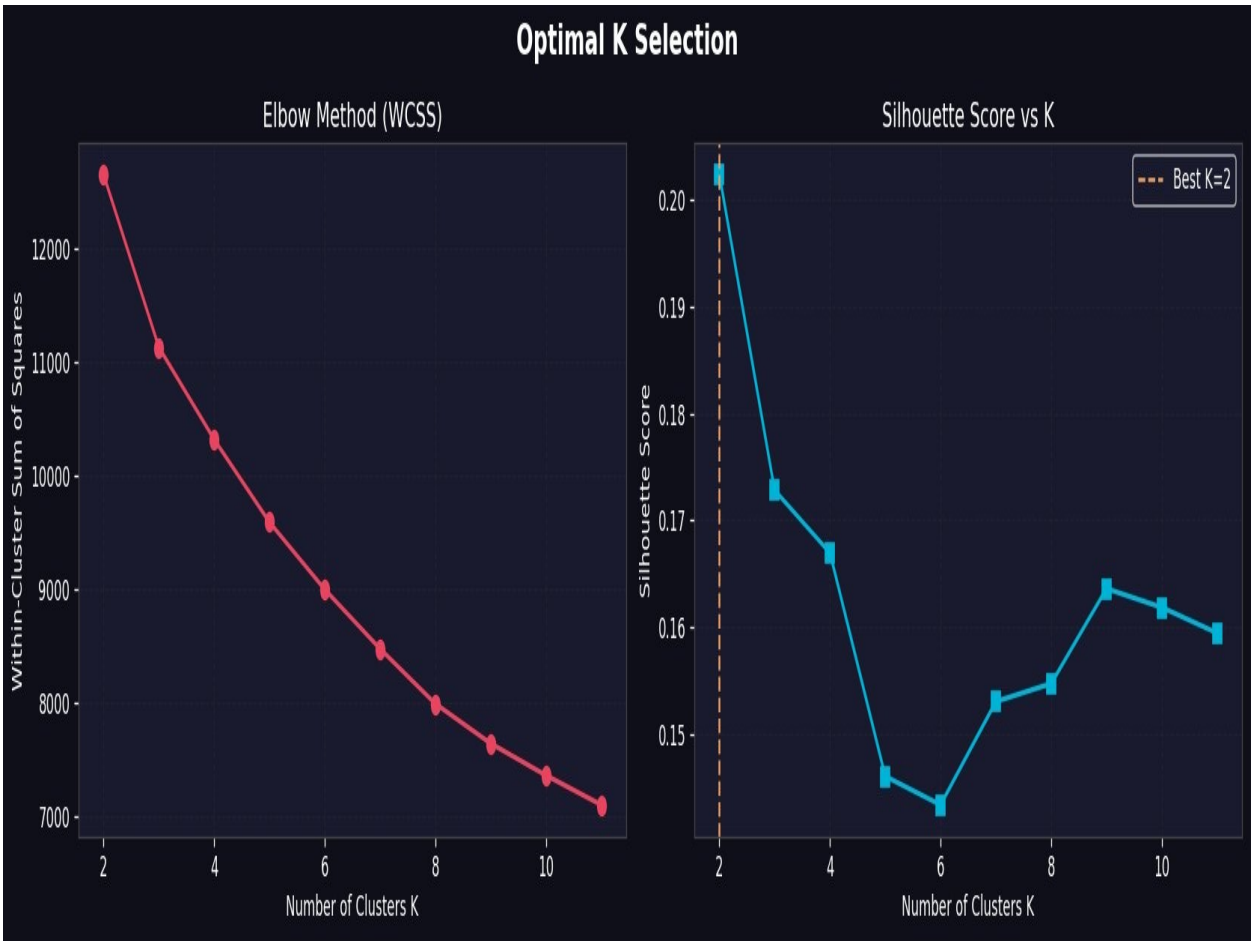

FIG. 5 — *Optimal K Selection — Elbow Method (WCSS) and Silhouette Score vs K; Best K* = 2.*

FIG. 6 presents the per-cluster silhouette coefficient bar chart at K* = 2. Both clusters C0 and C1 have profiles that rise steadily without abrupt negative dips, confirming reasonable within-cluster homogeneity.

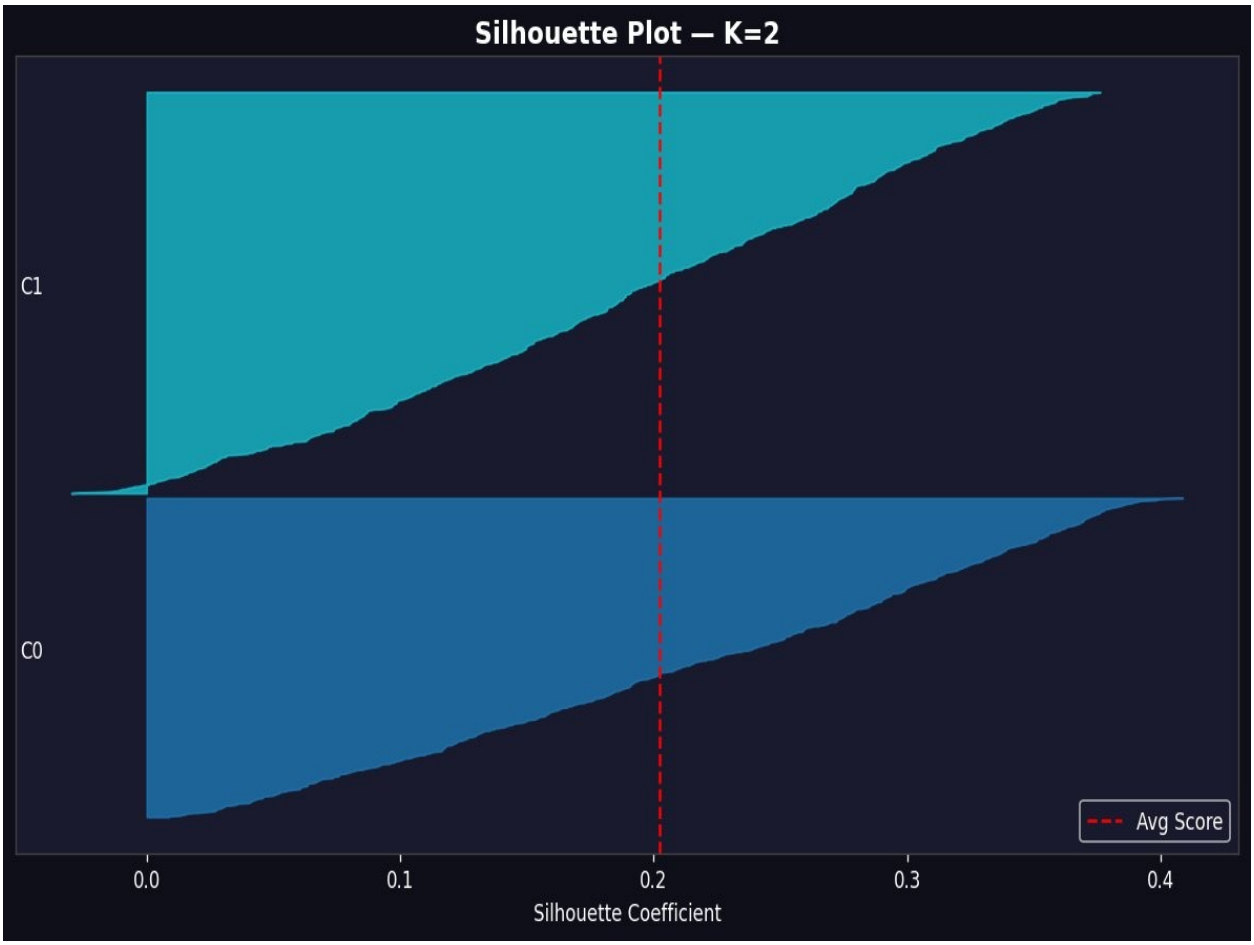

FIG. 6 — *Silhouette Plot at K* = 2.*

3.4 K-Means Clustering

K-Means partitions the feature space into K* Voronoi cells by iteratively updating cluster centroids to minimise total intra-cluster variance. The implementation uses the k-means++ initialisation strategy [6], which probabilistically selects initial centroids to reduce the expected number of iterations required for convergence. The algorithm is restarted ten times with independent initialisations; the solution with the lowest WCSS is retained.

3.5 Hierarchical (Agglomerative) Clustering

Agglomerative clustering begins with each data point in its own singleton cluster and iteratively merges the two clusters whose linkage distance is smallest. The system uses Ward's linkage criterion, which minimises the total within-cluster variance at each merge step. A dendrogram is computed on a random 200-track subsample for visual inspection, as shown in FIG. 7. The two macro-clusters emerge at a Ward distance of approximately 22.

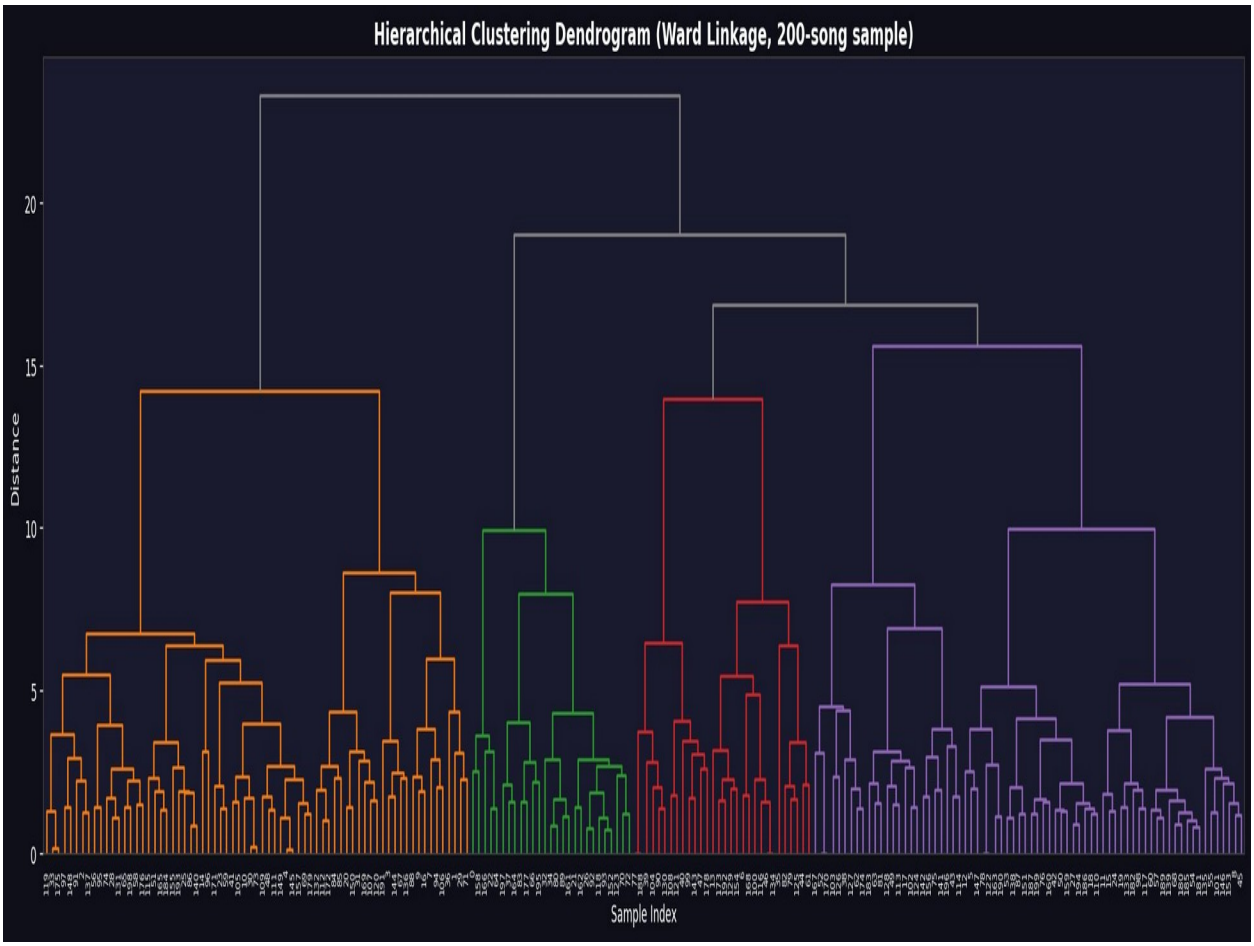

FIG. 7 — *Hierarchical Clustering Dendrogram (Ward Linkage, 200-track sample).*

3.6 DBSCAN Anomaly Detection

DBSCAN discovers clusters as contiguous high-density regions and categorises as noise any point that does not belong to a density-connected component. This property makes it the only algorithm in the pipeline that naturally identifies anomalous tracks. The neighbourhood radius eps is estimated from the elbow of the k-distance graph shown in FIG. 8 (k = 5). The elbow appears at approximately eps = 2.2.

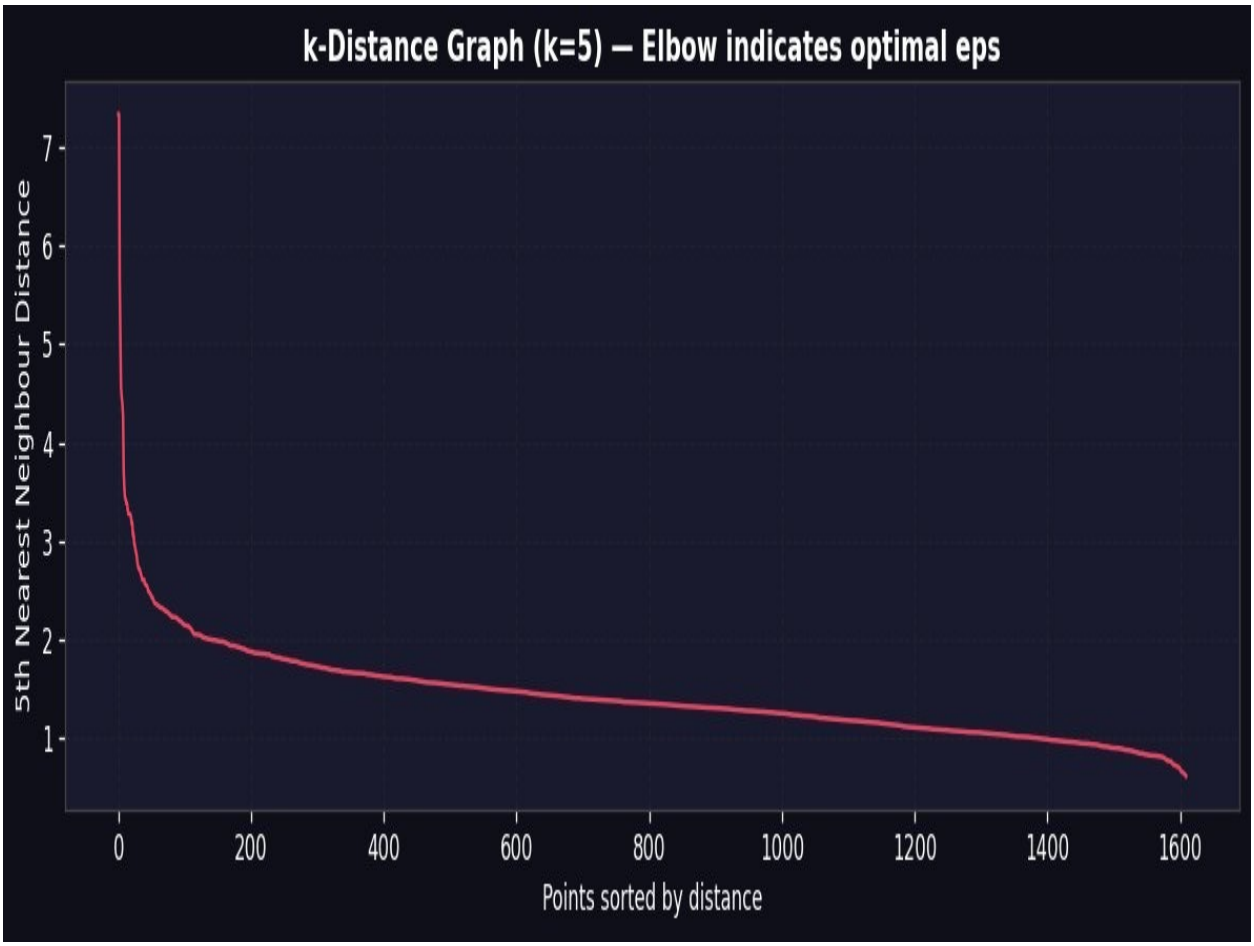

FIG. 8 — *k-Distance Graph (k=5) — Elbow indicates optimal eps for DBSCAN.*

3.7 Gaussian Mixture Models

GMM represents the dataset as a weighted sum of multivariate Gaussian distributions and assigns soft probabilistic membership weights to each track. Model order selection follows the Bayesian Information Criterion (BIC); the model minimising BIC over candidate component counts [2, 9] is selected. FIG. 9 shows both BIC and AIC curves. The BIC minimum occurs at 8 components; AIC continues to decline, suggesting further latent structure is present. Posterior probabilities are retained and assignments with maximum probability below 0.60 may optionally be flagged for manual review.

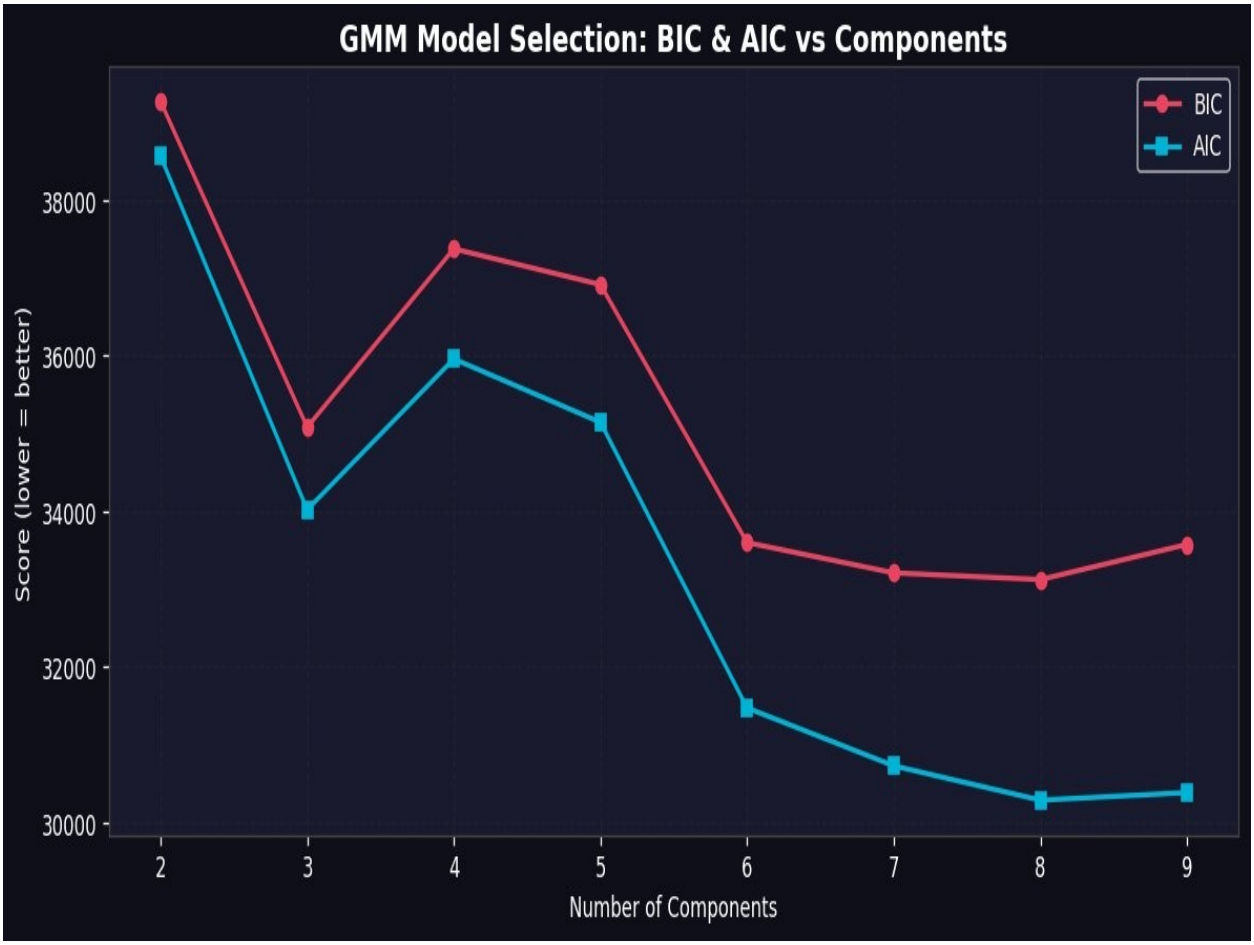

FIG. 9 — *GMM Model Selection: BIC & AIC vs Number of Components.*

3.8 Dimensionality Reduction and Visualisation

PCA projects the standardised feature matrix onto its first two and three principal components. In the experimental corpus, PC1 accounts for 30.3% of variance and PC2 for 16.5%, together preserving 46.8% of total variance. FIG. 10 and FIG. 11 show the 2D and 3D PCA scatter plots, respectively.

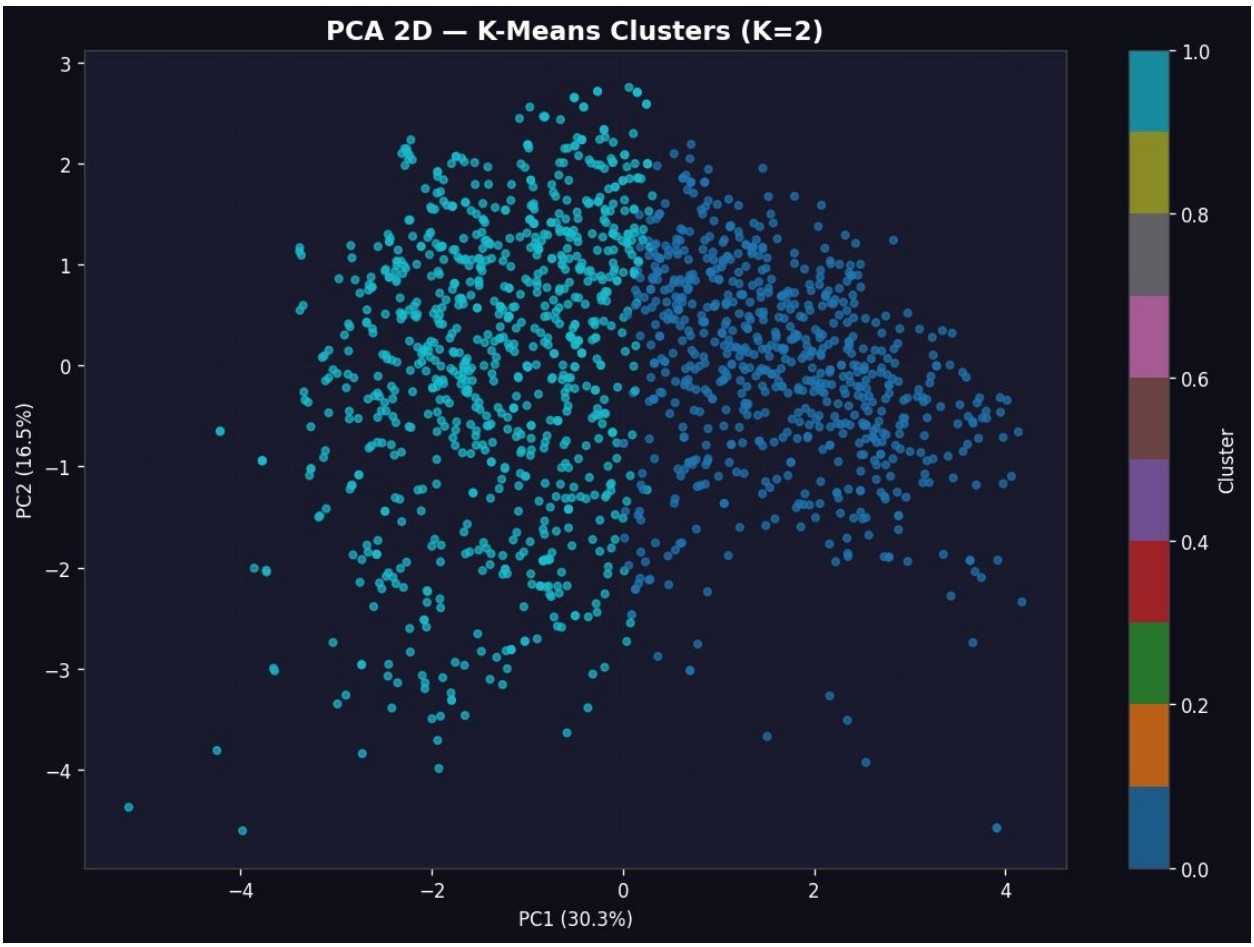

FIG. 10 — *PCA 2D Scatter — K-Means Clusters (K=2).*

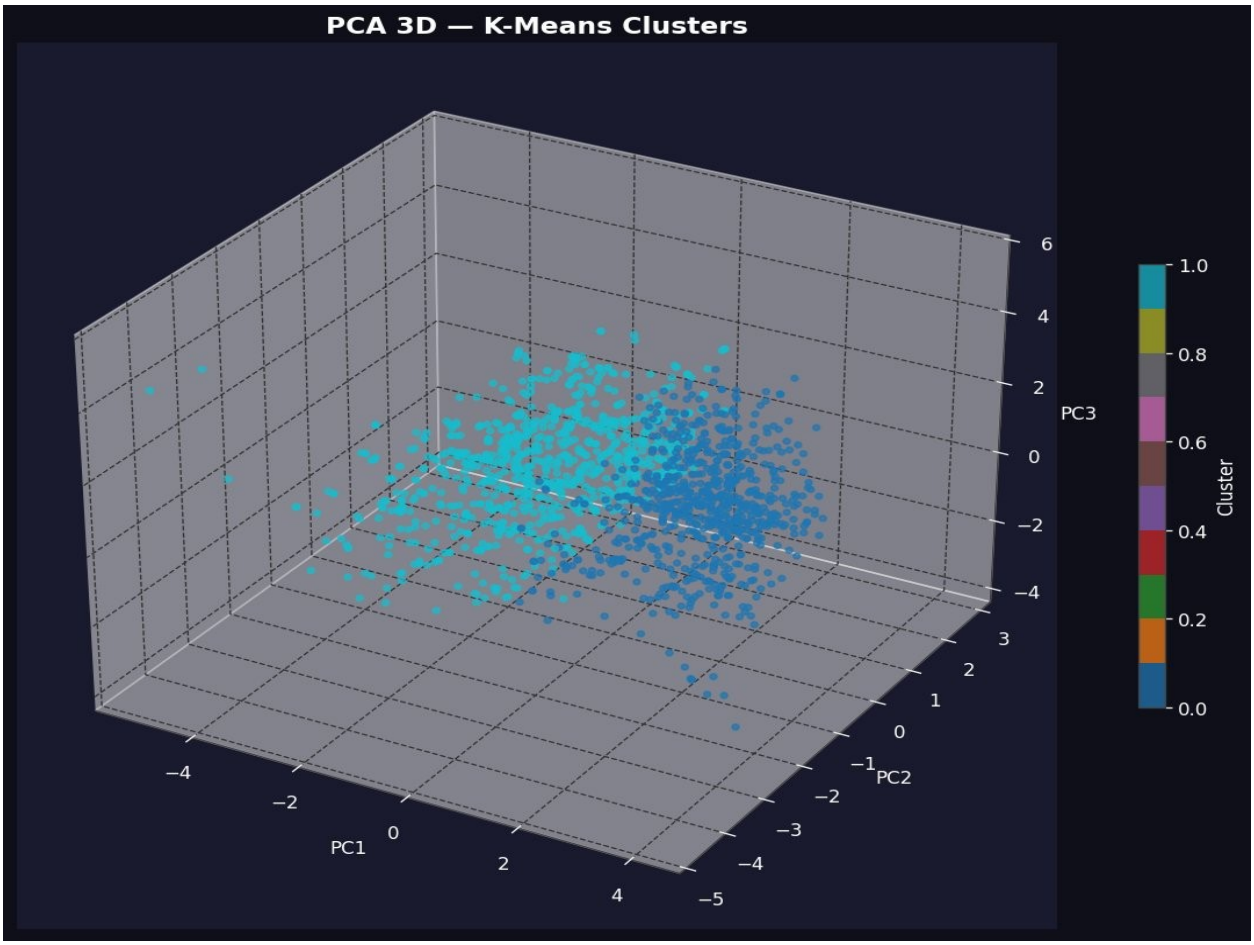

FIG. 11 — *PCA 3D Scatter — K-Means Clusters.*

t-SNE computes a non-linear two-dimensional embedding that preserves local neighbourhood structure [7]. FIG. 12 provides two t-SNE scatter plots: the left panel coloured by K-Means cluster label and the right panel coloured by Spotify popularity score. The popularity panel reveals that high-popularity tracks are not strongly spatially segregated within the feature space.

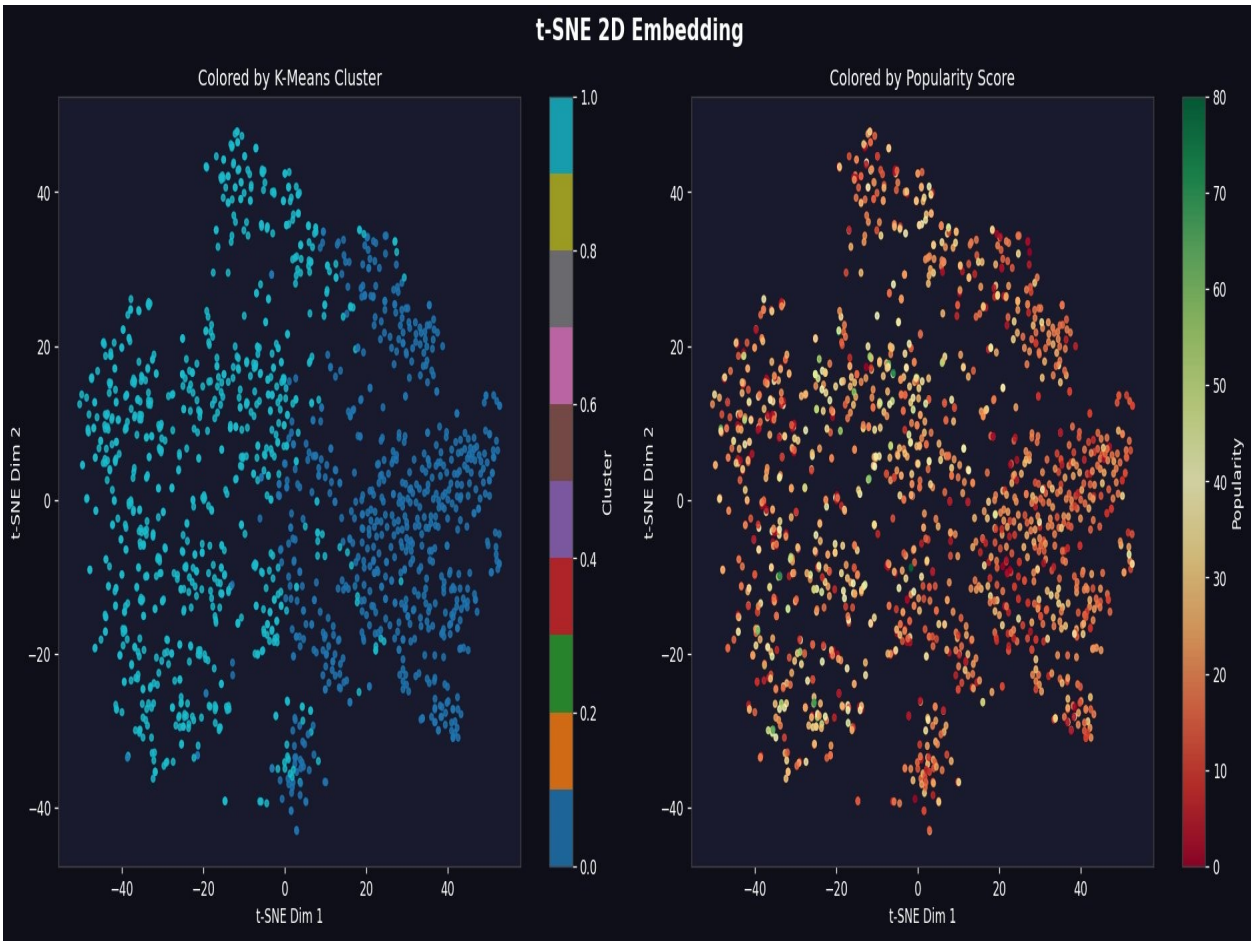

FIG. 12 — *t-SNE 2D Embedding — Coloured by cluster (left) and popularity score (right).*

3.9 Cluster Profiling and Musical Labelling

The system computes the within-cluster mean of every acoustic feature, producing a centroid table. Cluster names are assigned using a deterministic rule-based mapping: high energy with low acousticness yields "High-Energy Rock"; high acousticness with low energy yields "Acoustic & Mellow"; elevated danceability with positive valence yields "Groovy & Upbeat"; dominant instrumentalness yields "Instrumental / Experimental"; prominent liveness yields "Live Performance". In the K* = 2 solution, C0 (High-Energy Rock) maximises energy, liveness, loudness, speechiness, tempo, and duration, while C1 (Mid-Tempo Classic) leads on

acousticness, danceability, instrumentalness, and valence. FIG. 13 presents the normalised centroid heatmap and FIG. 14 the corresponding radar charts.

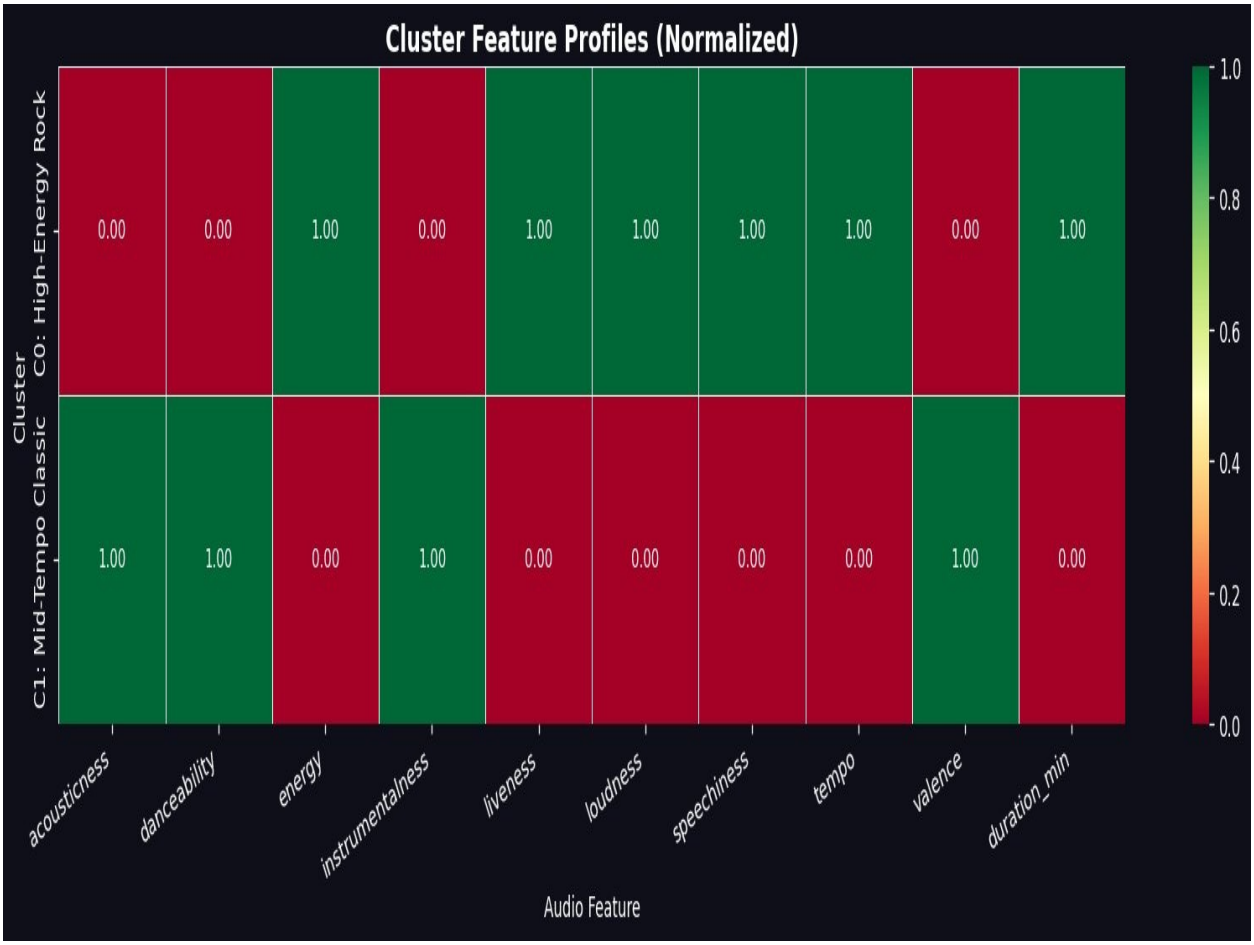

FIG. 13 — *Cluster Feature Profiles (Normalized) — Centroid heatmap.*

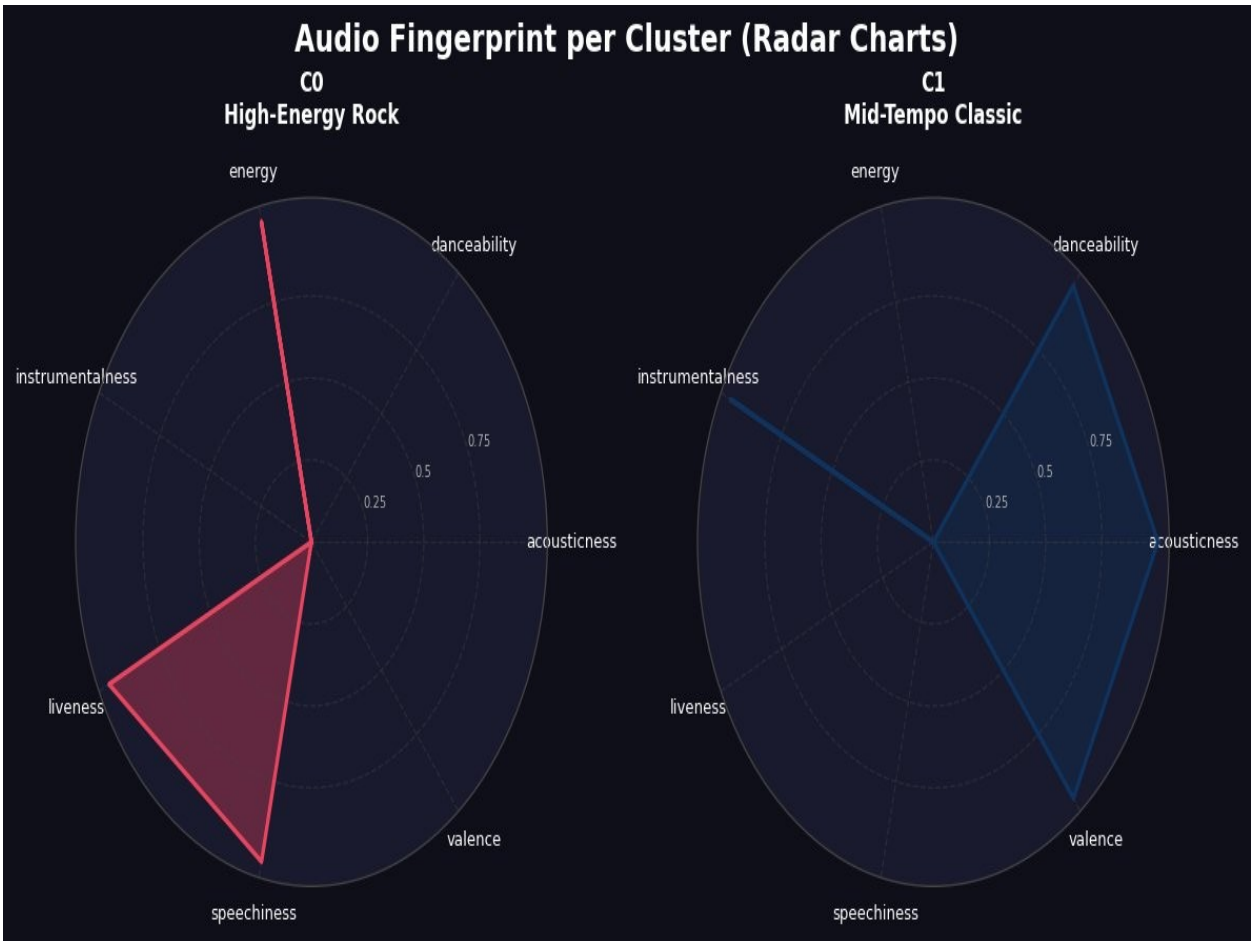

FIG. 14 — *Audio Fingerprint Radar Charts — C0: High-Energy Rock; C1: Mid-Tempo Classic.*

3.10 Temporal Analysis

The system computes a cross-tabulation of cluster assignment against recording decade. FIG. 15 presents grouped and stacked bar charts. The temporal analysis reveals a pronounced shift: tracks from the 1960s are overwhelmingly Mid-Tempo Classic (~92%), whereas by the 2000s–2010s the High-Energy Rock cohort grows substantially, reflecting the band's evolving sonic character.

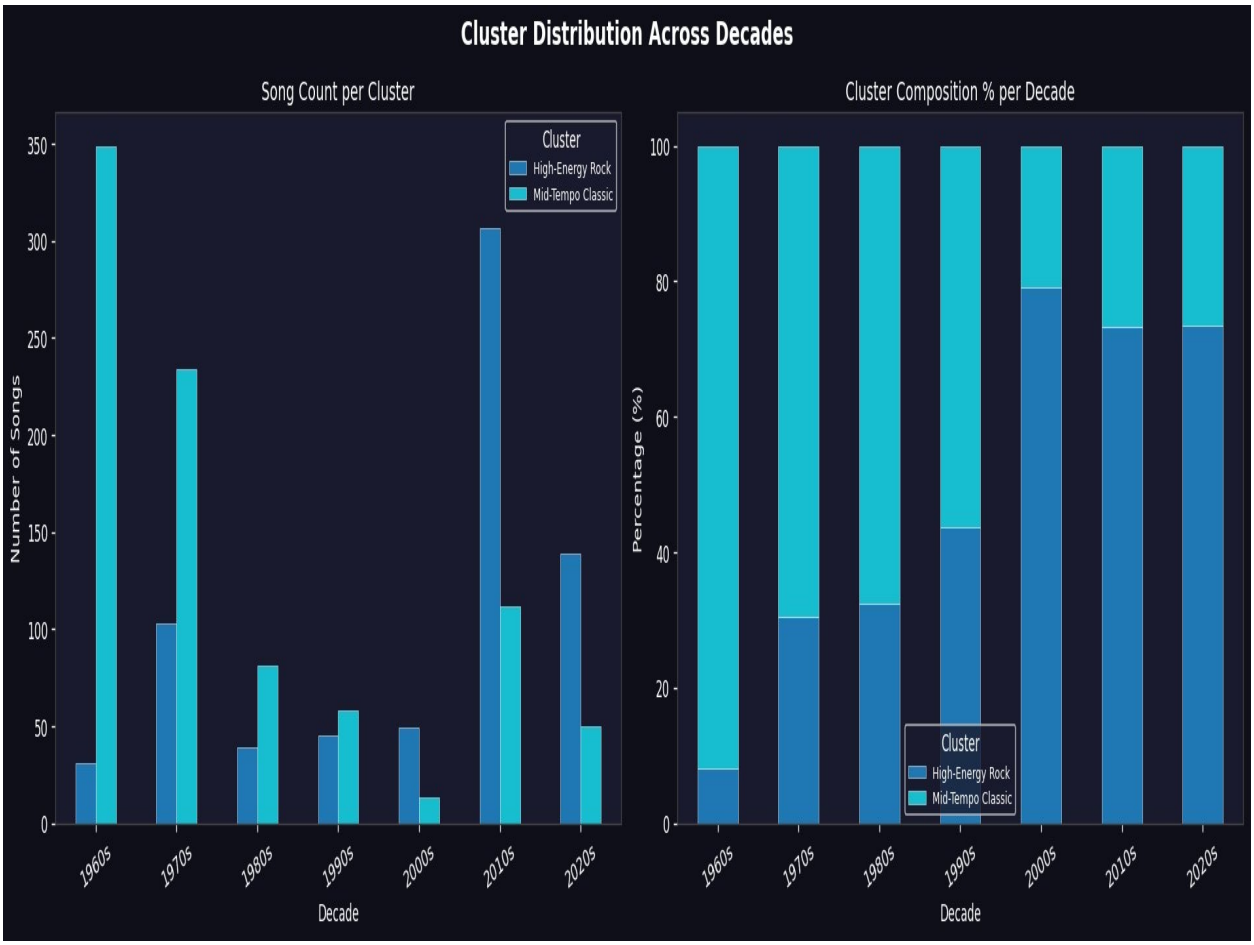

FIG. 15 — *Cluster Distribution Across Decades — Absolute count and percentage composition.*

3.11 Supervised Validation via Random Forest

A Random Forest classifier is trained to predict K-Means cluster labels from audio features on an 80/20 stratified train-test partition. Accuracy above the majority-class baseline confirms that the clusters have genuine internal consistency. Gini-based feature importance scores are extracted and ranked. FIG. 16 shows the results: liveness (0.327) is the dominant separator, followed by energy (0.164), danceability (0.147), and speechiness (0.138), together accounting for over 77% of discriminative power. Instrumentalness (0.012) and acousticness (0.020) contribute least.

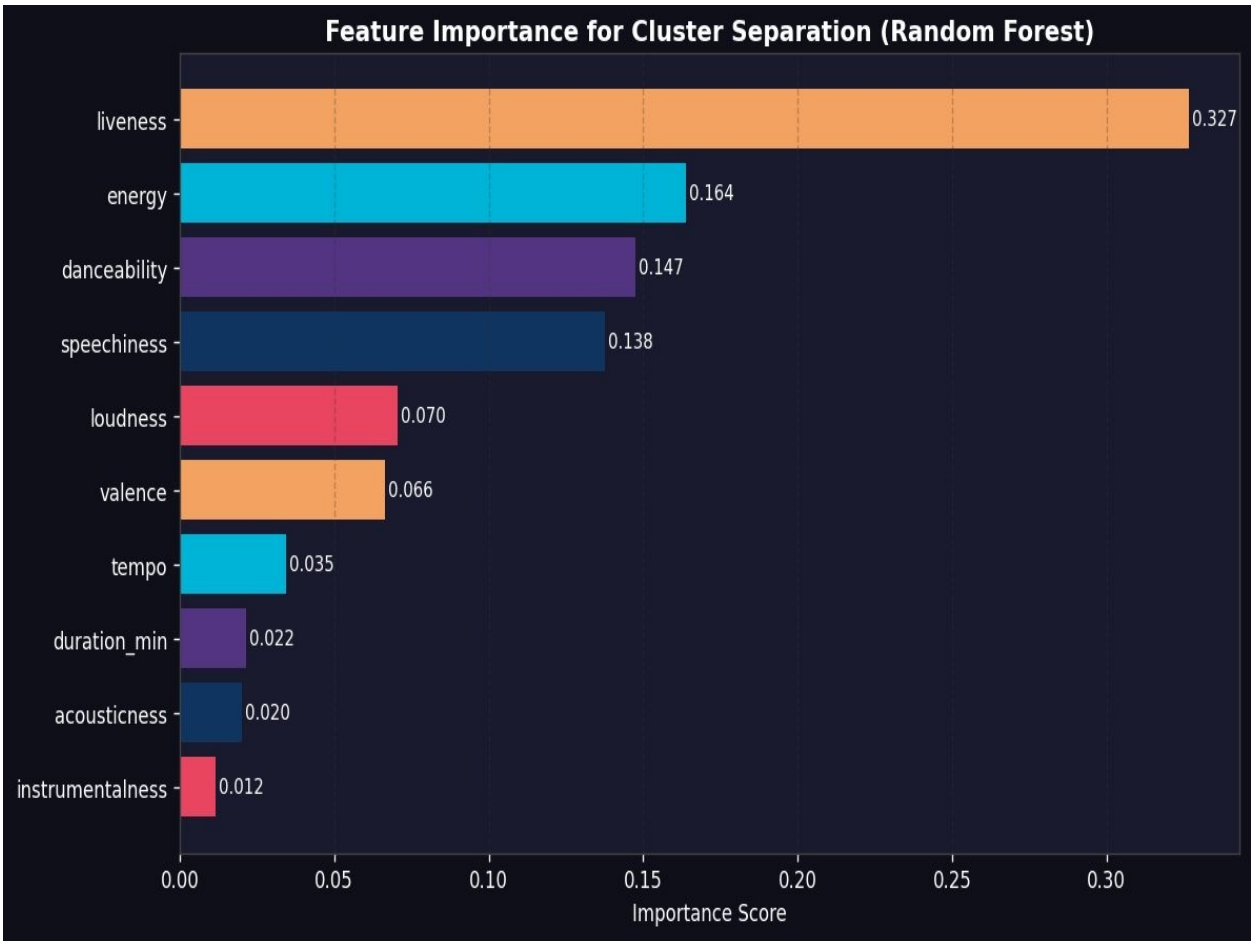

FIG. 16 — *Feature Importance for Cluster Separation (Random Forest).*

4. CLAIMS

What is claimed is:

1. A computer-implemented method for partitioning a music catalogue into cohorts, the method comprising: receiving a tabular dataset in which each row encodes one audio track and each column encodes a continuous acoustic feature; removing exact duplicate rows and rows containing missing values; applying zero-mean unit-variance standardisation to each acoustic feature column; determining an optimal cohort count K* by computing

within-cluster sum of squares and silhouette coefficient for candidate K values in [2, 11] and selecting the K yielding the highest mean silhouette coefficient; fitting a K-Means model with K-Means++ initialisation and ten independent restarts; assigning to each track a hard cohort label; and generating a centroid profile table and named label for each cohort using a deterministic rule-based mapping over centroid feature values.

2. The method of claim 1, further comprising fitting an Agglomerative Hierarchical Clustering model with Ward linkage and K* target clusters; computing a silhouette coefficient for the hierarchical solution; and generating a dendrogram visualisation on a randomly sampled subset of at least one hundred tracks.

3. The method of claim 1, further comprising fitting a DBSCAN model whose neighbourhood radius is estimated from the elbow of a k-distance graph; identifying as anomalous any track assigned the noise label (−1); and recording metadata attributes of anomalous tracks in a tabular report.

4. The method of claim 1, further comprising fitting a Gaussian Mixture Model for each candidate component count in [2, 9]; selecting the component count minimising the Bayesian Information Criterion; and computing probabilistic membership weights for each track.

5. The method of claim 1, further comprising projecting the standardised feature matrix onto two and three principal components via PCA; projecting the matrix onto two dimensions via t-SNE with configurable perplexity; and generating scatter plots coloured by cohort label and by track popularity score.

6. The method of claim 1, further comprising training a Random Forest classifier to predict cohort labels from audio features on an 80/20 stratified train-test partition; extracting Gini-based feature importance scores; and ranking acoustic features by their contribution to cohort separability.

7. The method of claim 1, further comprising deriving a calendar decade attribute from the track release date; computing a cross-tabulation of cohort labels against recording decade; and generating stacked and grouped bar visualisations showing temporal cohort composition.

8. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the processors to perform the method of any of claims 1 through 7.

9. A system comprising one or more processors and a memory storing a music cohort discovery pipeline, wherein the pipeline implements data ingestion, standardisation, multi-algorithm unsupervised clustering, dimensionality reduction, cluster profiling, anomaly detection, and supervised validation as recited in claims 1 through 7, and wherein the pipeline produces machine-readable cohort assignments and human-readable visualisation artefacts as outputs.

5. BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a histogram and box plot illustrating the popularity distribution of the 1,610-track corpus.

FIG. 2 is a lower-triangle Pearson correlation heatmap for all ten acoustic features.

FIG. 3 is a bar chart showing track counts per recording decade.

FIG. 4 shows violin plots for all ten acoustic feature distributions.

FIG. 5 shows the Elbow Method (WCSS) and Silhouette Score curves used to select K* = 2.

FIG. 6 is a silhouette coefficient bar chart at K* = 2.

FIG. 7 is a Ward-linkage dendrogram on a 200-track random subsample.

FIG. 8 is a k-distance graph (k = 5) used to estimate the DBSCAN neighbourhood radius eps.

FIG. 9 shows BIC and AIC curves versus number of GMM components.

FIG. 10 is a two-dimensional PCA scatter plot coloured by K-Means cluster label.

FIG. 11 is a three-dimensional PCA scatter plot coloured by K-Means cluster label.

FIG. 12 shows t-SNE 2D embeddings coloured by cluster label (left) and popularity (right).

FIG. 13 is a normalised centroid heatmap for clusters C0 and C1.

FIG. 14 shows radar charts depicting the audio fingerprint of each named cohort.

FIG. 15 shows grouped and stacked bar charts of cluster composition per recording decade.

FIG. 16 is a horizontal bar chart of Random Forest feature importance scores.

6. REFERENCES CITED

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ABSTRACT OF THE DISCLOSURE

Recorded music catalogues have grown to an enormous scale, and the manual organisation of tracks into meaningful groups has become practically unfeasible. This invention presents a computational method and software system that automatically discovers natural cohorts within a music catalogue by analysing multi-dimensional audio feature representations extracted from streaming platform metadata. Four unsupervised learning algorithms — K-Means, Agglomerative Hierarchical Clustering, DBSCAN, and Gaussian Mixture Models — are integrated within a unified pipeline that includes principled model selection, dimensionality reduction via PCA and t-SNE, and a supervised validation stage using Random Forest feature importance. Experiments on 1,610 Rolling Stones tracks demonstrate musically meaningful groupings including High-Energy Rock and Mid-Tempo Classic cohorts. Applications include automated playlist generation, recommendation engine pre-processing, music archiving, and audio discovery interfaces.